

A PYTHON-BASED INTRUSION DETECTION SYSTEM

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01 Introduction

A Python-based Intrusion Detection System is a security tool designed to monitor network traffic and detect suspicious activities such as port scanning, packet flooding, TCP flag scans, suspicious port usage, and ARP spoofing indicators. This project uses Python with Scapy for packet capture and analysis, and a graphical interface to display alerts, dashboards, logs, and testing evidence in real time.

02 Problem Statement

Manual network monitoring is time-consuming and difficult for beginners. Existing IDS tools such as Snort, Suricata, and Zeek are powerful, but they can be complex to configure and interpret. Therefore, a simple Python-based IDS is needed to detect suspicious network behaviour and present alerts in a more understandable, beginner-friendly way.

03 Objectives

- To capture live packets or analyse PCAP files.
- To extract key traffic features
- To detect suspicious network activities.
- To generate simple IDS alerts.

04 Technical Methodology

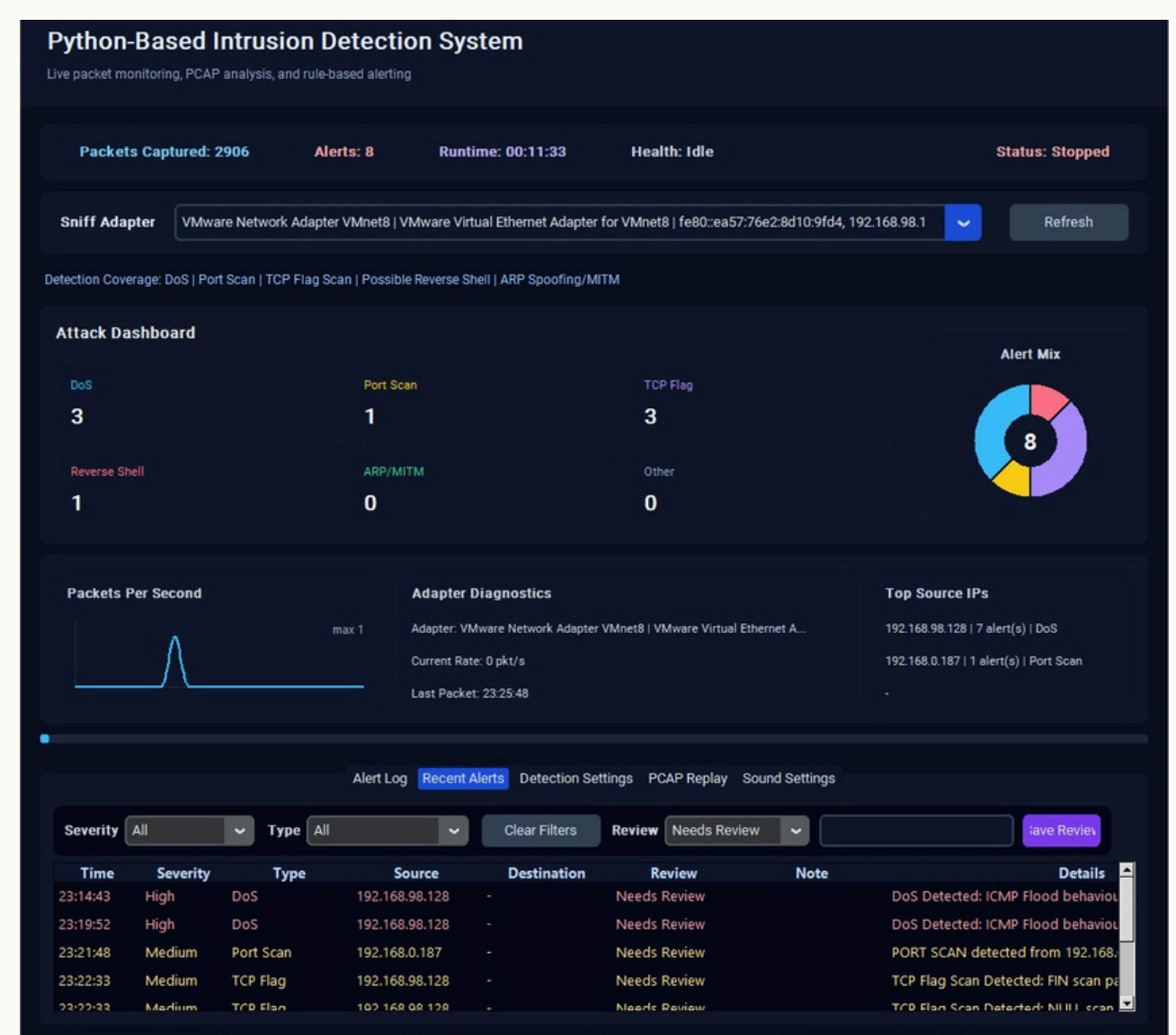
The IDS was developed using Python with a graphical dashboard for live packet monitoring and PCAP analysis. The system captures packets from the selected network adapter or loads saved PCAP files. Packet details are then processed by the detection engine to identify suspicious behaviour such as DoS attacks, port scans, TCP flag scans, possible reverse shell activity, and ARP spoofing/MITM. When a threat is detected, the system displays alerts, updates the dashboard, and allows results to be exported as CSV or summary reports.

05 System Features

- Live packet capture and PCAP analysis
- Rule-based detection for DoS, scans, suspicious ports, and ARP spoofing
- Real-time dashboard with graphs, alerts, and source tracking
- Adjustable thresholds and detection rule toggles
- Alert review, sound notification, and export support

06 Results & Discussion

The developed IDS successfully monitored network traffic and detected multiple suspicious activity indicators during testing. As shown in Figure 1, the system captured 2,906 packets and generated 8 alerts, including DoS-style ICMP Flood, Port Scan, TCP Flag Scan, and Possible Reverse Shell Indicator alerts. The dashboard also provides real-time visualization, alert logs, source IP tracking, and attack statistics.



07 Conclusion

The Python-based IDS achieved its objective of providing a beginner-friendly alternative to complex network monitoring tools. It can capture traffic, analyse packets, and alert users to suspicious indicators such as DoS-style behaviour, port scans, TCP flag scans, suspicious port usage, and ARP spoofing. Overall, it serves as an accessible educational tool for understanding foundational network security and rule-based intrusion detection.

